PROBLEM SOLVING THROUGH PROGRAMMING (CSE103)

Question Bank- 2025

UNIT 1:Introduction To Computer and C language, Operators and Expressions	
	Theory Questions
1	Describe the essential steps involved in the process of developing a computer program.
2	List the primary components of a computer system and explain the role of each in information processing.
3	What are the key rules to follow when naming variables in C programming? Give two valid and two invalid examples of variable names.
4	List and Explain important features of C programming language
5	Define algorithm and flowchart. Write an algorithm and flowchart to find the largest of 3 numbers.
6	Explain any 4 input and output devices.
7	Write the basic structure of a C program and explain each section using an appropriate example.
8	Explain the significance of different basic data types in C with examples.
9	Illustrate the following: keywords, tokens, constants
10	Explain the following operators with example: a) Relational operators b) Logical operator c) Bitwise operators
11	Define type conversion in C. Explain its types with suitable examples.
12	Describe formatted and unformatted I/O operations in C with examples.
	Explain the usage of conditional operator in C. Evaluate the following expressions: (i) Let x = 3, y = 4, z = 4 c = (z >= y >= x ? 100 : 200) (ii) Let c=5, d=0, e=10 a = c > 1 ? d > 1 || (e = 0) ? 100 : 200 : 300
13	Evaluate the following expression (related expressions) i. X = a – b / 3 – c * 2 – 1 when a=9,b=12, c=3 ii. 10 != 10 || 5 < 4 && 8

	i. 100 % 20 <= 20 – 5 + 100 % 10 – 20 == 5 >= 1 != 20 ii. a += b *= c -= 5 where a=3, b=5 and c=8 Iii. (5 + 3) * 2 - 8 / 4.
14	Differentiate clearly between the following operators with suitable C code: Pre-increment, Post-increment,Pre-decrement, Post-decrement
	programs
1	Write a C program to compute both Simple Interest and Compound Interest based on user inputs.
2	Write a C program to perform all basic arithmetic operations (addition, subtraction, multiplication, division, and modulus) on two user-input integers. Display the results clearly.
3	Write a C program using logical operators to check if a number lies between 1 and 100.
4	Write a C program to compute the area of a circle and perimeter of a circle.
5	Write a C program to compute the area of a circle and perimeter of a triangle.
6	Write a C program that computes the size of int, float, double and char variables
7	Develop a C program to find the largest of three numbers using a ternary operator.
8	Write a C program to swap two numbers using temporary variable
UNIT- 2:Decision control and Looping statements	
	Theory Questions
1	Explain the syntax and flowchart of the if statement, else if and if-else-if ladder statement in C.
2	Describe nested –if statement in C with syntax for checking largest of three numbers.
3	Outline the structure and syntax of the switch-case control structure in C. Also explain the significance of the break statement in switch.
4	Define a loop? Explain for,while and do-while loops with a syntax and example.
5	Compare while and do-while loop.
6	What are unconditional statements? List them and explain
7	Explain goto statement with an example.
8	Summarize break and continue statements with examples

	Explain the roles of argc and argv in command line arguments.
	Programs:
1	Write a program to check whether a given number is even or odd.
2	Write a C program using a switch statement to assign grades based on marks (entered as an integer from 1 to 5), where 1 → A 2 → B 3 → C 4 → D 5 → F Any other input → Invalid Choice
3	develop a C program that determines whether an input character is a vowel or consonant using switch statement.
4	Write a program that prints all numbers between 1 and 50 that are divisible by 7 using a for loop.
5	Write a C program using the else-if ladder to calculate the electricity bill based on the number of units consumed using the following slab: 0–50 units: ₹3/unit 51–150 units: ₹5/unit 151–250 units: ₹7/unit Above 250 units: ₹10/unit
6	Write a C program to simulate a simple calculator that performs arithmetic operations using switch statement. Error message should be displayed, if any attempt is made to divide by zero.
7	Develop a C program to find the factorial of a number using while/do-while/for loop
8	Develop a C program to generate fibonacci series using while/do-while/for loop.
9	Develop a C program to find the sum of n numbers OR to find sum of the series: 1+2+3+.....+n.
10	Check whether the number is palindrome or not.
11	Find whether a year is a leap year or not.

	Write a C program to find the sum of all the digits and occurrence of a digit in the number.
	Write a C program to find the GCD and LCM of two numbers using Euclid‘s method.
12	Design a C program that accepts two numerical values and an arithmetic operator (+, -, *,/) as command-line arguments. The program should compute and display the result of the specified operation.
UNIT-3 Arrayas and strings	
	Theory Questions -
1	Define Strings. How do you declare and initialize a string in C? Explain with example.
2	Write a program to read and display a string using the gets() and puts() functions.
3	Explain the following string manipulation functions with programming example. strlen(), strrev(),strcat(),strcmp(),strcpy()
4	What is an array? Explain with an example how to declare and initialize a One-dimensional array.
5	Explain with an example how to declare and initialize a Two-dimensional array.
6	What is the difference between a one-dimensional and a two-dimensional array in C?
	Programs
1	Write a C program to implement linear search technique.
2	Write a C program to search for a key element using binary search technique.
3	Write a C program that reads N integer numbers and arrange them in ascending order using selection Sort.
4	Write a C program to sort the array elements using bubble sort.
5	Write a program to find the sum of two matrices.
6	Write a program to multiply two matrices.
7	Write a C program to search for an element in a sorted array using binary search.

8	Write a C program to transpose a matrix
9	Write a program to reverse a string without using strrev().
10	Write a program to find whether two strings are anagrams using the strcmp() function.
11	Write a program to check if a given string is a palindrome
12	Write a program to find the length of a string without using strlen().
13	Write a program to concatenate the string without using strcat()
14	Write a program to count the number of vowels in a given string.
	UNIT-4 Pointers, Functions
	Theory Questions
1	Define a pointer. List the benefits of using pointers in C
2	Explain how pointers are declared and initialized with general syntax and example
3	What is a function? Explain different classification of user defined functions based on parameter passing and return type with examples.
4	Explain function call, function definition and function prototype with examples.
5	What are actual parameters and formal parameters? Illustrate with example.
6	Illustrate the concept of 'scope of a variable' in c with an example program.
7	What is the difference between call by value and call by reference?
8	What is meant by recursion? Give an example of its usage.
	Programs
1	Write a C program to read N integers into an array A and find the sum of elements using pointers.
2	Write a C program using pointers to perform the following: i. display the address of a variable ii. display the value stored at the address
3	Write a program in C using functions to swap two numbers using global variables concept and call by reference concept.

4	Write a recursive function to calculate the factorial of a number.
5	Write a recursive function to generate Fibonacci sequence up to n terms.
6	Write a program to find the largest number among three integers using a function.
7	Write a recursive function in C to check whether a string is a palindrome.
8	Implement a program that merges two sorted arrays into one sorted array using functions.
9	Write a c-program using functions to generate the Fibonacci series.
	UNIT-5- Structures, Files
	Theory Questions
1	What is a Structure? Explain the syntax of a structure with an example.
2	Differentiate between array of structures and structure of arrays.
3	Which are the two ways in which structure variables can be created? Explain with examples.
4	Define a file. List and explain the functions used in C to perform basic file operations
5	Write the general syntax of fopen() and fclose() functions. Explain the different file opening modes.
	Programs
1	Write a program to store and display details (name, age, marks) of 5 students using an array of structures.
2	Write a program to pass a structure (student with marks in 3 subjects) to a function and calculate the average marks.
3	Create a structure called Employee with the fields: Name, Branch and Experience and input the details of two employees.
4	Write a program to copy contents of one file into another.
5	Write a program that reads characters from a file, counts the frequency of each vowel, and displays the count.
6	Write a C program to copy contents of one file to another file.

7	Write a C program to perform reading and writing from a file using fprintf() and fscanf().